Pranav Saxena

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Education

University of Edinburgh — Edinburgh, UK
MSc by Research Informatics (Supervised by Prof. Hakan Bilen) — *Sept 2026 – Aug 2027*

Birla Institute of Technology and Science, Pilani — Goa, India
Bachelor of Engineering in Electronics and Communication — *Oct 2022 – May 2026*

Experience

Visiting Researcher — Jun 2025 – Aug 2026
Carnegie Mellon University — *Pittsburgh, USA*

- Pursued my undergraduate thesis at the AirLab in the Robotics Institute, under the supervision of Dr. Wenshan Wang and Dr. Ji Zhang (onsite Jun–Dec 2025, remote since Jan 2026).
- Worked on 3D Object Representation Learning and Vision-Language Navigation for Outdoor Scenarios, resulting in LLM-RG (IROS 2025 workshop). Organized the CMU-VLA Challenge as part of IROS 2025.

Research Intern — May 2024 – Dec 2025
National University of Singapore — *Remote*

- Research intern at MARMoT Lab, under the supervision of Prof. Guillaume Sartoretti and co-supervised by Prof. Marija Popović (TU Delft) on active 3D reconstruction.
- Worked on projects spanning 3D Scene Graph Generation using VLMs, Open-Vocabulary Gaussian Splatting, and Next-Best-View Planning for NeRFs, leading to ZING-3D (ICRA 2026 workshop) and Gen-LangSplat (preprint).

Publications and Preprints

Pranav Saxena, Nishant Raghuvanshi, Neena Goveas. *"UAV-VLN: End-to-End Vision-Language Guided Navigation for UAVs"*. European Conference on Mobile Robots 2025 (Oral). [Paper]

Pranav Saxena, Avigyan Bhattacharya, Ji Zhang, Wenshan Wang. *"LLM-RG: Referential Grounding in Outdoor Scenarios using Large Language Models"*. Human-Aware Embodied AI Workshop, IROS 2025. [Paper]

Pranav Saxena, Jimmy Chiun. *"ZING-3D: Zero-Shot Incremental 3D Scene Graphs via Vision-Language Models"*. Multimodal Embodied Interaction in Robots Workshop, ICRA 2026. [Paper]

Pranav Saxena. *"Gen-LangSplat: Generalized Language Gaussian Splatting with Pre-Trained Feature Compression"*. preprint. [Paper]

Projects

Persistent Object Identity and Tracking Through State Transformations — Sept 2026 – Present
University of Edinburgh

- Developing an online, multi-object tracking framework for egocentric videos that follows objects through state transformations.
- Maintaining persistent object identity through these changes with a structured, state-aware memory for every object.

3D Object Representation Learning — Dec 2025 – Present
Carnegie Mellon University

- Learning generalized 3D object representations from point clouds and multi-view visual observations for downstream 3D understanding and reconstruction tasks.
- Developing object-level shape completion methods to reconstruct complete, geometrically consistent point clouds from partial observations during SLAM.

Vision Language Navigation for Outdoor Scenarios | Paper, Code — Jun 2025 – Nov 2025
Carnegie Mellon University

- Developed a zero-shot hybrid referential grounding pipeline for outdoor driving scenarios.

- Leveraged off-the-shelf Vision-Language Models (VLMs) for detailed visual understanding and Large Language Models (LLMs) for symbolic reasoning, without end-to-end retraining.
- Achieved 64.72% and 67.91% accuracy on val and test subsets of the Talk2Car dataset, respectively.

3D Scene Graph Generation using VLMs | Paper Apr 2025 – Sept 2025

National University of Singapore (NUS)

- Developed a zero-shot pipeline for incremental 3D scene graph generation using Vision-Language Models (VLMs), enabling open-vocabulary reasoning and geometric grounding.
- Achieved 97% node labels accuracy on scenes in the Replica dataset and 96% on scenes in the HM3D dataset.
- Subsequently extended the framework for object-goal navigation tasks.

Efficient 3D Open Vocabulary Gaussian Splatting | Paper, Code Dec 2024 – May 2025

NUS, TU Delft

- Developed a generalized framework for efficient language-grounded 3D Gaussian Splatting eliminating per-scene autoencoder training and improving cross-scene generalization.
- Implemented a cross-scene generalized autoencoder trained on ScanNet for CLIP feature compression into a 16-dimensional latent space.
- Achieved 84.4% localization accuracy on the LERF dataset and 93.3% mIoU on the 3D-OVS dataset, with a nearly 2× reduction in training time.

Next-Best-View Policy for NeRFs | Code May 2024 – Nov 2024

NUS, TU Delft, Purdue University

- Developed an attention-based Next-Best-View (NBV) policy for Neural Radiance Fields to select the most informative viewpoints for 3D reconstruction.
- Implemented a Vision Transformer (ViT) that encodes multi-view image embeddings and candidate poses to predict ΔPSNR gain, guiding view selection.
- Integrated the framework with pixelNeRF to compute ground-truth ΔPSNR via differentiable rendering and supervise policy training under limited measurement budget.

Technical Skills

Languages: Python, C/C++, SQL
Frameworks: ROS, ROS2, PyTorch, TensorFlow, CUDA, Keras, Flask
Tools: Git, Docker, Arduino IDE, RViZ, Gazebo

Teaching and Reviewing

Tutor and Marker, Discrete Mathematics and Probability, University of Edinburgh

Lab Demonstrator, Machine Learning Practical (Postgraduate), University of Edinburgh

Reviewer, IEEE Transactions on Robotics (T-RO), ACM Journal on Autonomous Transportation Systems (JATS)

Honors and Awards

SAE Aerothon'24, Nationwide Best Mission Award Nov 2024

Founded Team Artemis, BITS Pilani-Goa's autonomous drone team, and led the automation subdivision. Developed autonomous capabilities for aerial surveying, object detection, and precision payload delivery. Ranked 3rd among 100 teams in Phase I and won Best Mission Award in the final round.

Indian Olympiad Qualifier in Astronomy May 2022

Qualified Part-1 (NSEA). Among top 264 students in India to be shortlisted for Part-2 (INAO). Received Certificate of Merit from Indian Association of Physics Teachers.

Leadership

Organizer, CMU-VLA Challenge

Lead organizer of the CMU-VLA Challenge 2025, held as part of IROS 2025. Managed the overall organization and evaluation of the challenge.